

LEO ZHOU

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Education

Rice University

August 2025 – December 2028

Bachelor of Science in Computer Science

Houston, Texas

GPA: 3.92/4.00 **Relevant Coursework:** Data Structures & Algorithms, Linear Algebra, Computer Organization, Discrete Math, Intro to Computer Engineering, Probability and Statistics for Data Science, Multivariable Calculus

Technical Skills

Languages: Python, C/C++, Swift, JavaScript/TypeScript, SQL (PostgreSQL), HTML/CSS

Web & Backend: React, Node.js, FastAPI, Flask, REST APIs, WebSockets, Redis (Pub/Sub)

Infrastructure: Docker, GitHub Actions (CI/CD), Git, Locust, Sentry

ML / Edge AI: PyTorch, TensorFlow Lite Micro, Quantization (PTQ/QAT), Knowledge Distillation, Pruning

Embedded & Mobile: STM32 (bare-metal), FreeRTOS, ESP32, UART/SPI/I²C, SwiftUI, Apple Vision (OCR)

Spoken Languages: English (native), Mandarin (native), Spanish (conversational)

Experience

CanSemi

June 2026 – Present

Software Engineer Intern

Guangzhou, China

- Built event-driven binary parser extracting parametric test records from 500+ ATE datalogs, replacing manual regex parsing
- Designed normalized PostgreSQL schema (lot/wafer/die/result) & batch pipeline loading 40 lots (~12,000 die) in <2 min/lot
- Developed a FastAPI backend with 8 query endpoints serving wafer/lot-level test data, replacing manual reporting
- Built an interactive React/Canvas wafer map rendering 300+ die with hover and drill-down, replacing static Excel heatmaps

ABC Reads

May 2026 – Present

Founder & iOS Engineer

Remote

- Founded a native SwiftUI iOS app for heritage Chinese speakers, personalized by a per-user Known-Character Model
- Built a real-time camera pipeline (Apple Vision OCR + word segmentation) overlaying pinyin on only unfamiliar characters
- Compiled a 120,000-entry offline SQLite dictionary via a Python pipeline, frequency-ranked against a SUBTLEX-CH corpus
- Designed a local-first data layer over the raw SQLite3 C API powering spaced-repetition review and a WidgetKit widget

Rice Wind Energy — Controls Subteam

October 2025 – April 2026

Firmware Engineer

Houston, Texas

- Developed ESP32 firmware in C/C++ for a 3-phase wind turbine controller with RPM, power, and pitch-control sensing
- Built a 6-state safety state machine with fault latching/recovery and a perturb-and-observe peak-power tracking algorithm

A-Lab@Rice

November 2025 – February 2026

Undergraduate Researcher

Houston, Texas

- Built COMSOL Multiphysics simulations for a solar thermal desalination project (STREED) to optimize model efficiency
- Executed parameter sweeps and analyzed outputs to improve model performance and support design alterations

Projects

SeatLive | Python, FastAPI, PostgreSQL, Redis, React, WebSockets, Docker, GitHub Actions

May 2026 – June 2026

- Built a real-time booking engine preventing double-booked seats via atomic conditional UPDATES with Redis TTL holds
- Implemented idempotent confirmation via client-supplied keys, short-circuiting retries instead of double-processing
- Proved correctness with CI concurrency tests firing 50 simultaneous requests at one seat, asserting exactly one succeeds
- Streamed live seat-map updates via WebSockets + Redis pub/sub; Locust load test served 4,459 requests with zero failures

Edge-ML Audio Compression Benchmark | PyTorch, TensorFlow Lite Micro, C++

June 2026 – July 2026

- Built a PyTorch compression pipeline (distillation → INT8 quantization → pruning) for a 50-class audio classifier
- Compressed a 3.9 MB CNN teacher to 71 KB (55× smaller) at 70.8% vs. 77.5% teacher accuracy on a \$5 microcontroller
- Benchmarked PTQ vs. QAT on a Pareto frontier; distillation-aware QAT recovered FP32 accuracy at 3.1× compression
- Deployed the INT8 model to an ESP32-S3 with TFLite Micro and a custom C++ mel-spectrogram front end, achieving ~300 ms on-device inference (10× real time) in ~300 KB RAM

STM32 Bare-Metal Motor Controller | Embedded C, STM32, FreeRTOS

March 2026 – July 2026

- Wrote bare-metal C firmware on a Nucleo-F401RE with HAL-free GPIO, UART, SPI, I2C, and timer drivers from scratch
- Implemented closed-loop PID speed control via PWM and quadrature encoder feedback with <2% steady-state error
- Ported the control task to FreeRTOS; built a UART telemetry pipeline with Python scripts plotting PID step response

Leadership / Extracurricular

Junior National Fencing Team Captain | Fencing New Zealand

November 2023 – May 2024

- Captained NZ's junior national team to Oceania titles; competed at Junior Commonwealth Games & World Championships
- Earned 27 national first-place finishes; coached 100+ youth fencers and refereed hundreds of bouts across Oceania